

10 (a) (i) Magnetic force provides centripetal force. M1

$$Bqv = mr\omega^2 = mr\left(\frac{2\pi}{T}\right)^2 \quad A1$$

$$T = \frac{2\pi m}{qB}$$

(ii) Into the page/plane B1

$$(iii) f = \frac{qB}{2\pi m} = \frac{(2 \times 1.60 \times 10^{-19})(0.850)}{2\pi(6.88 \times 10^{-27})} \quad M1$$

$$= 6.29 \times 10^6 \text{ Hz} \quad A1$$

(iv) Gain in KE per revolution
 $= 2 \times 450 \times 2 \times 1.60 \times 10^{-19}$ M1

$$= 2.88 \times 10^{-16} \text{ J} \quad A1$$

(v) Gain in KE = $\frac{1}{2}mv^2$
 $3 \times 2.88 \times 10^{-16} = \frac{1}{2}(6.88 \times 10^{-27})(v^2)$ M1

$$v = 501000 \text{ m s}^{-1} \quad A1$$

(vi) • Same time intervals when travelling with constant speed B1
 • (Shorter and shorter time intervals when accelerating)

• Smaller increase in speed each acceleration B1

(vii) • The cyclotron uses the same accelerating gap, which enables it to become compact and therefore saves cost in many ways. B1

• The accelerating particles can be taken to higher energy states within the small space available.

• Besides the small size of a cyclotron, there are other factors which reduce the cost of this device. For example the cost of the building which contains it, the cost of the foundation and the cost of radiation shielding.

• Another big advantage of a cyclotron is that it has only one electric driver. This makes it cost-efficient because of less cost equipment and less costly power.

• Cyclotrons can produce the particle beam in a continuous form.

(b) (i) $E = -B/v = -(0.080)(1.5)(3.00)$
 $= 0.36 \text{ T}$ M1

$$I = \frac{E}{R}$$

$$= \frac{0.36}{4.0} = 0.090 \text{ A} \quad A1$$

(ii) b. Induced e.m.f. flows from lower to higher potential. B1

(iii) When the rod is moving at constant speed, due to the cutting of magnetic flux, there will be an induced current through it. The B1

direction of the induced current will be such that there will be a magnetic force to oppose the motion causing it. B1

Therefore, to keep the rod moving at constant speed, a force needs to be applied

- (iv) Work done by external force = Electrical energy supplied to keep charges moving / heat dissipated by current in the resistor R / joule heating in resistor R B1
- (v) $F = BIL$
= 0.080 (1.5)(0.090) M1
= 0.0108 N A1